

Steel, Lining & Filling Support

26/08/2019

Steel Support I

- *Steel is an expansive material for support.
- *Steel is free from natural defects.
- *Steel has high Young's modulus of elasticity.
- *Steel is least effected by temperature & humidity.
- *Steel can be reused.
- *Steel consists of “H” section steel member.
- *This section is used to prepare different curvature.
- *It has advantages if installation with wase & speed.
- *Support system is reliable & has high load carrying capacity.

Steel Support II

- *Steel set needs little maintenance, if any.
- *Steel sets has high cost & difficulty in adopting in varying ground condition.
- *Steel sets needs more space in comperiasam to shot-creating & rock bolting.
- *Steel set is used as permanent support fpr mine entries.

Rigid Steel Arches I

- *Rigid steel arches are used as permanent support in mines for permanent workings.
- *Rigid arches are composed of two, three or four segments arches, forming near semicircular design.
- *Arch shape is more efficient use of steel sections than flat cross bars.
- *Steel members in arches are in compression instead of bending.

Yielding Steel Arches I

- *Yielding steel arches are composed of three sections.
- *Top section slides between two side elements.
- *After regular interval (fort night) tightening elements are loosened, arch sliders, converges & thus relived stress accumulated on them.
- *This steps eliminates deformation.
- *In some design, yielding can be achieved by inserting wooden pieces between steel elements.

Steel Tubing I

- *Cast iron steel tubing are used to permanently line shaft wall during shaft sinking.
- *Tubbing is commonly used when shaft is sunk with freezing method.
- *There are two available designs of tubbing, Germans & English tubbing.
- *cast iron tubing is built up on cast iron ring.
- *Each segments of ring compromises a number of flanged segments shaped to suit curvature of shaft.

Rock Bolt I

- *Rock bolt has largely replaced timber as support.
- *underground excavation has become safer.
- *Underground space is not restricted by timber.
- *Ventilation of place is not obstructed by timber.

Rock Bolt Theory I

- *Beam theory works for layered or stratified deposit when rock bolts are used to support them.
- *Concept is that by bolting immediate roof acts as a beam to support over lying strata.
- *Suspension theory states that using roof bolts, immediate roof is suspended to main roof, which is strong & stable.
- *Rock bolts also reinforce rock by pressure arch & support discrete blocks.
- *Rock bolts are used in mine extensively.

Rock Bolt Theory II

- *Rock bolts are used as permanent support to roof, side wall, road ways, road ways junction, wide chambers area, stoping area
- *Rock bolts are used at brows, draw points & all other openings.
- *Number of rock bolts per square meter is called bolt density.

Slot & Wedge Rock Bolt I

- *This rock bolts are oldest, most simple & easy to manufacture & install.
- *Rock bolts consists of a steel rod with a slot cut at one end, which contains steel wedges.
- *At beginning wedge is placed in slot.
- *Wedge is driven in to slot by hammering exposed end or rock bolts.
- *Expanding halves grip hole wall, allowing bolts to be put under tension & carry load.
- *Blast vibration & ground movement easily loosened wedge type rock bolts.

Slot & Wedge Rock Bolt II

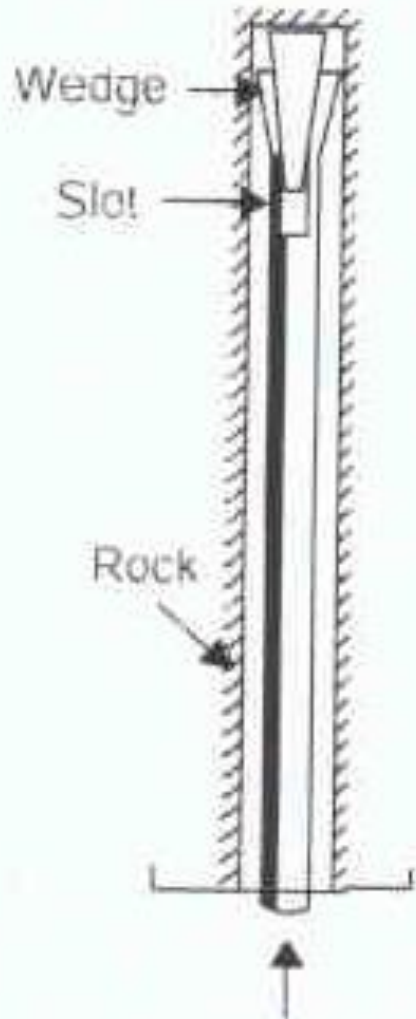
- *Holes are drilled with suitable drill diameter.
- *Hole length is 5 to 7 cm less than rock bolt rod length.

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Slot & Wedge Rock Bolt I



(a) Hammer in rockbolt
slot & wedge bolt

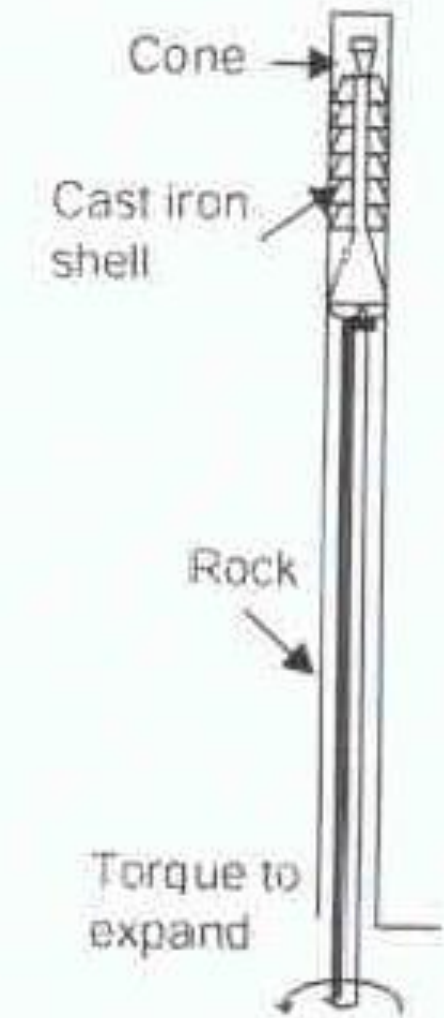
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Expansion Shell Bolt I

- *it consists of toothed blades of malleable cast iron with a cast iron conical wedge at one or both ends.
- *One or both cones are internally threaded on rock bolts, so a wrench cone is forced in to blades to press against wall of drilled hole.
- *Grip increases as tension increases.
- *These rock bolts are less expensive & very widely used for short term support in underground mines.
- *In soft rock, this rock bolt tends to slip & gets loosened.
- *Few days after installation, nuts are to be tightened as it gets loose with time initially for few days.

Expansion Shell Bolt I

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olt (b) Expansion
shell bolts

Cement Grouted Bolt I

- *In fully grouted rock bolt, grout rod does not have any mechanical anchor.
- *Rock bolt consist of a ribbed reinforcing bar, installed in a drill hole & bonded to rock over full length with cement mortar.
- *Grouted rock bolts are self tightening when rock starts to move & dilated.
- *Grouted rock bolt should be installed as soon as possible after excavation before rock has started to move & before it lost its interlocking & shear strength.

Cement Grouted Bolt II

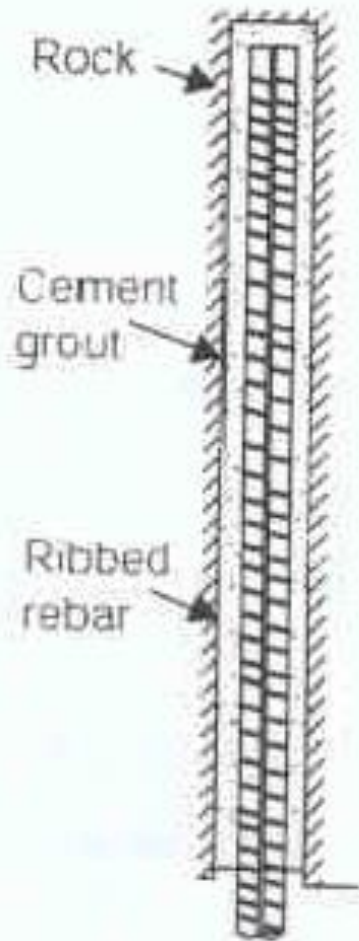
- *Normal grout mixture consist of sand, cement ratio as 50/50 or 60/40 ratio.
- *Water to cement ratio should not be greater than 0.4 by weight.
- *Common cement grout mixture ratio is 1:1:0.5, cement : sand : water by volume.
- *Exact proportion for particular raw material is fixed by hit & trial at field.
- *Grout injection, particularly in up hole, requires to ensure complete filling.
- *Some time, air pockets may be left in hole & this is difficult to detect.

Cement Grouted Bolt III

- *Pneumatically operated grout pump / loader are used to fill holes to which bar is driven.
- *Rock bolt is retained in up holes by placing a stone / widen chip at hole mouth.
- *Rod can also be retained in hole by packing drill hole collar with cotton waste, steel wool or wooden wedges.
- *Cotton waste is used at hole mouth to prevent fall of cement mixture from drill holes.
- *As rod is being pushed in cement filled holes, some excess cement mixture comes out & to be removed.

Cement Grouted Rock Bolts I

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(c) Cement grouted
dowel Rock bolts
different types

Cement Grouted Bolt IV

*Slight bend is given to rod of rock bolt insertion in hole, so that rock bolt does not come out from hole of their own.

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Rock Bolt Calculation I

Beam theory for layered deposit

$$B = (0.75qL) / (bh_1 + bh_2)$$

$B_c = (0.75qL) / b(h_1 + h_2)$ h_1 & h_2 are binded together

$$B_c / B = 1/2$$

B = Magnitude of bending stress, layers are not binded together

B_c = Magnitude of bending stress, layers are composed (binded together)

q = uniform load applied on width

L = width of opening

b = Width of rock slab

Rock Bolt Calculation II

Length of Rock bolt,

$$l = h * 1.05 \##$$

l = length of rock bolt

h = thickness of immediate roof in m

$\##$ = factor varies depending roof condition

Rock Bolt Calculation III

Number of rock bolts,

$$m \geq (LhcD)/R$$

m= no of rock belts

L= drive (gallery) width in m

h= thickness of immediate roof in m

c= distance between rows of bolts in m

D= immediate roof rock density

R= allowable axial force in ton

Rock Bolt Calculation IV

Allowable axial force,

$$R \geq (0.785d^2y)/n$$

R= allowable axial force in ton

d= diameter of rock bolt in mm

y= yielding strength of steel in ton/m sq

n= factor of safety

Rock Bolt Calculation V

Bolt density,

$$Bd = m/(Lc)$$

Bd= bolt density

m= no of rock bolt

L= drive (gallery) width in m

c= distance between rows of bolt in m

Rock Bolt Calculation VI

Bolt spacing,

$$b = L/m$$

b = bolt spacing

L= drive (gallery) width in m

m= no of rock bolt

Rock Bolt Calculation 2

l = length of rock bolt

h = thickness of immediate roof in m

m = no of rock bolt

L = drive (gallery) width in m

c = distance between rows of bolts in m

D = immediate roof rock density

R = allowable axial force in ton

d = diameter of rock bolt in mm

y = yielding strength of steel in ton/m sq

n = factor of safety

Rock Bolt Calculation I

Bd= bolt density

b = bolt spacing

Cable Bolting I

- *Cable bolts (grouted cable) are used in stopes to support back, to prevent its fall on caving.
- *Horizontal or near horizontal cable bolt also stabilizes & prevents caving of hang wall
- *Cable bolt can be of 18m or more, in length.
- *Up ward cable bolts stabilizes crown pillar in open stope
- *Cable bolts helps to increases stope span & reduce pillar dimensions.
- *Cable bolt can eliminate need of any other type of support.

Cable Bolting II

- *Used & old haulage / winder rope is widely used for cable bolting.
- *Before using old rope to use as cable bolt, all lubricants, grease in rope is to be removed.
- *Cable bolting can be used to stitch a weak flat roof or back.
- *This application is known as roof stitching or side wall stitching.
- *Roof / side wall stitching often accompanied with shotcrete.

Perfo Bolts I

- * Trawl in to two halves of a split perforated sleeve, is filled with cement grout by pumping.
- * Two halves of perforated trawl are placed together & bounded at end with soft iron weir.
- * Tube filled with cement grout is inserted in drilled hole.
- * Cement grout is not placed directly in drilled hole.
- * Rock bolt (dowel) is then hammered by sledge hammer in side cement filled trawl.
- * Cement grout is then flow out of perforated tube filling up drill hole.

Perfo Bolts II

*Perfo bolt with cement grout fill can be installed in hole with out help of compressed air.

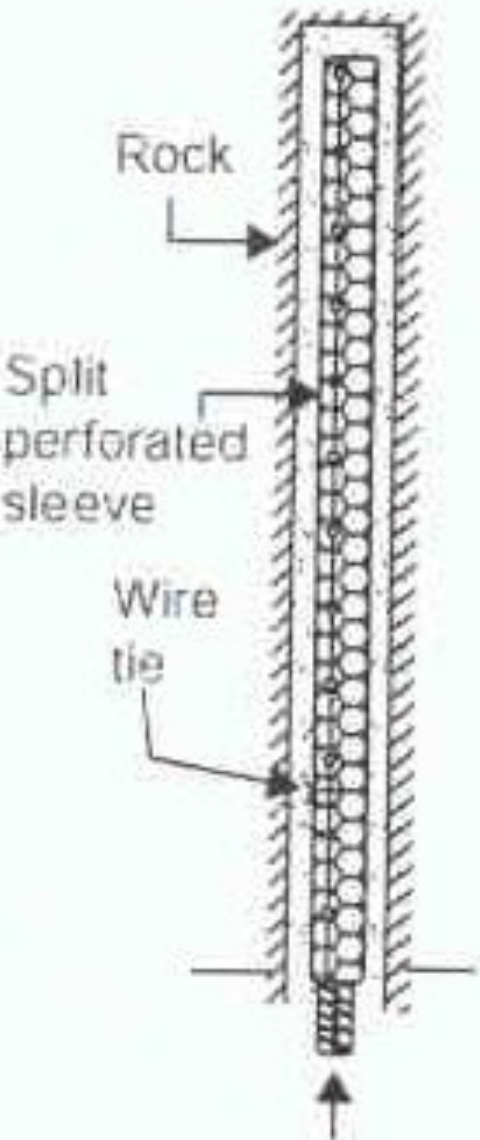
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Perfo Bolts I

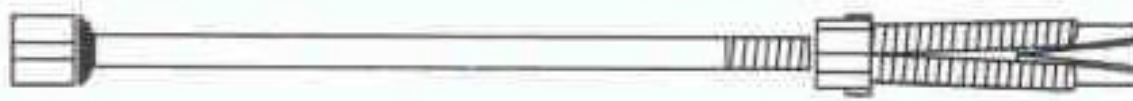


(d) Hammer in rebar perfo-bolt

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Rock Bolts I

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Mechanical bolt



Swellex



Split set

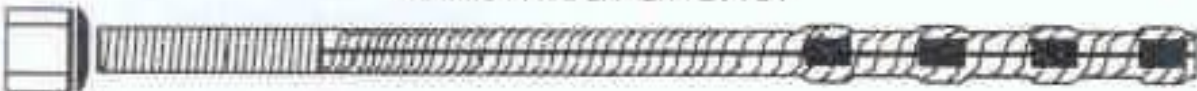
Friction bolts



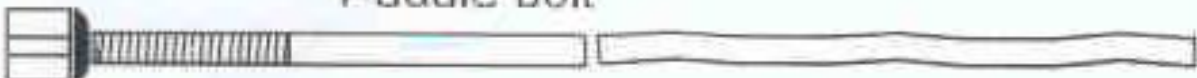
Posimix



Chemical anchor



Paddle bolt



Wrigle bolt

Resin bolts

Resin Grouted Bolt I

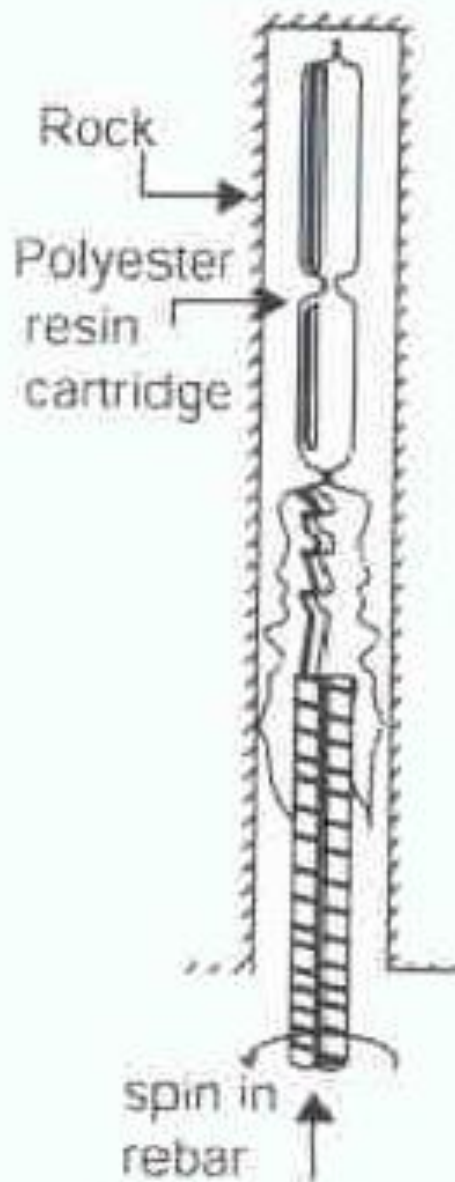
- *Resin grouted bolt is used where high & quick strength is required.
- *Resin grouted is costly.
- *A ribbed reinforcing rod is cemented in to drill holes is cemented in to drill holes by a polyester resin.
- *Drill hole is filled with resin cartridge & rock bolt rod is inserted in drilled hole.
- *Material inside resin cartridge changes in to thick liquid & than to a high strength solid by process of catalysts-initiated polymerization.

Resin Grouted Bolt II

- *Resin grouted rock bolts takes full strength within 2-4 hours.
- *Bond strength is much more than that of cement grouted bolt.
- *Exact setting time for rock bolt can be designed by inserting slow setting cartridge in hole.
- *In case resin sets before complete installation of rock bolts, then rock bolt remains protruded out of drill hole.
- *This rock bolt shall be ineffective.

Resin Grouted Bolt I

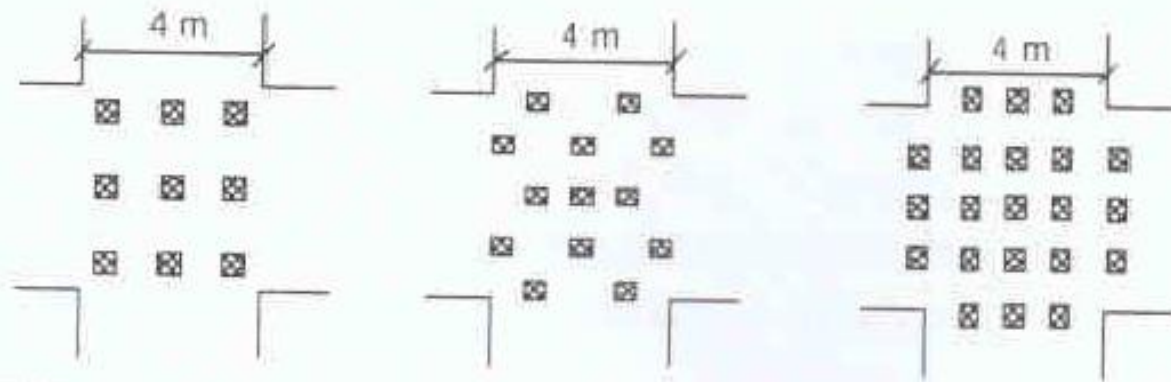
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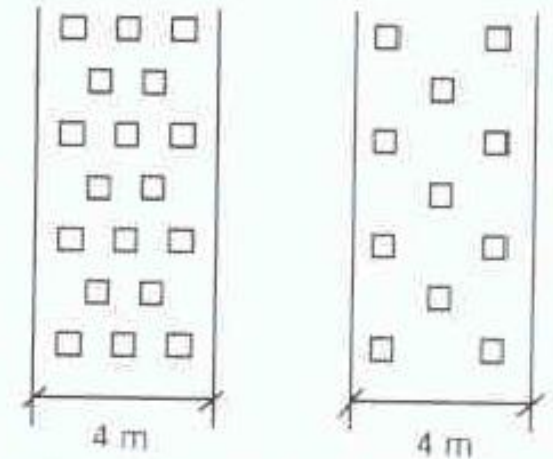
(e) Resin-bonded bolt

Use of Microwave I

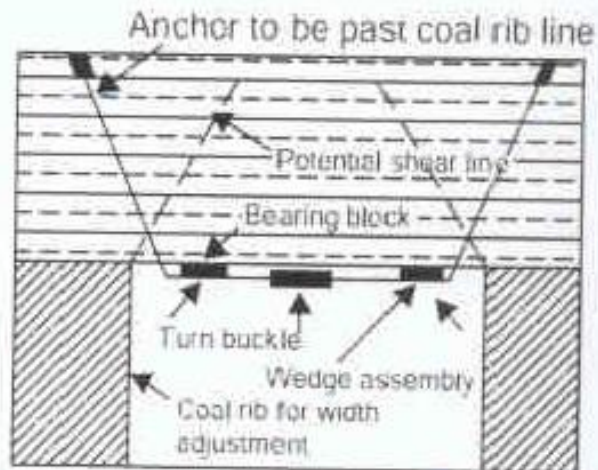
different types



(f) Supporting 4-way junctions



(g) Supporting roadways/galleries



(h) Roof stitching

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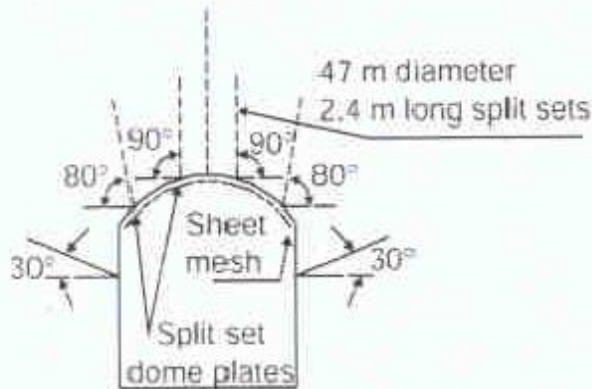
Wire Mesh I

- *Wire mesh is also known as screen.
- *Wire mesh is available in different wire gauge, thickness & aperture.
- *Function of wire mesh is to support rock in between rock bolts.
- *Wire mesh is imparted when rock is closely jointed & bolts are moderately to widely spaced.
- *Wire mesh also serves as reinforcement for shotcrete.

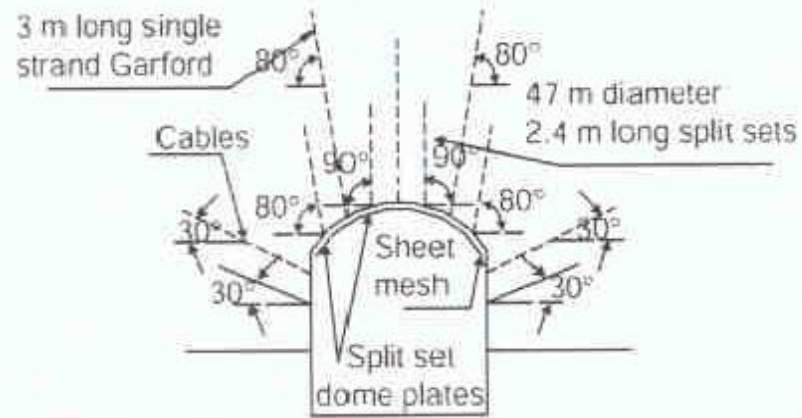
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Wire Mesh I

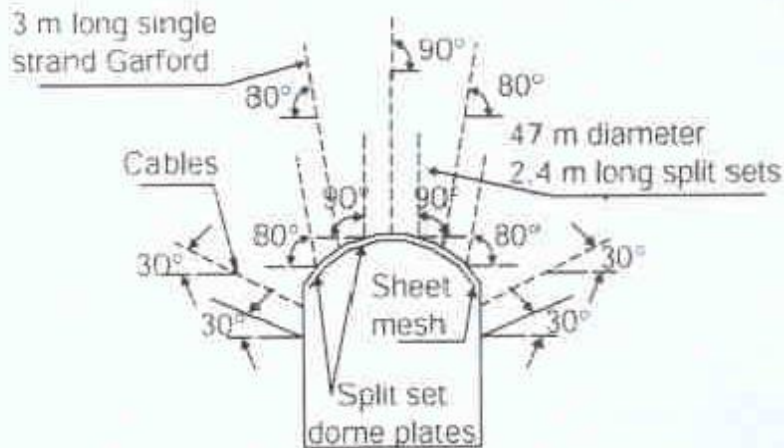
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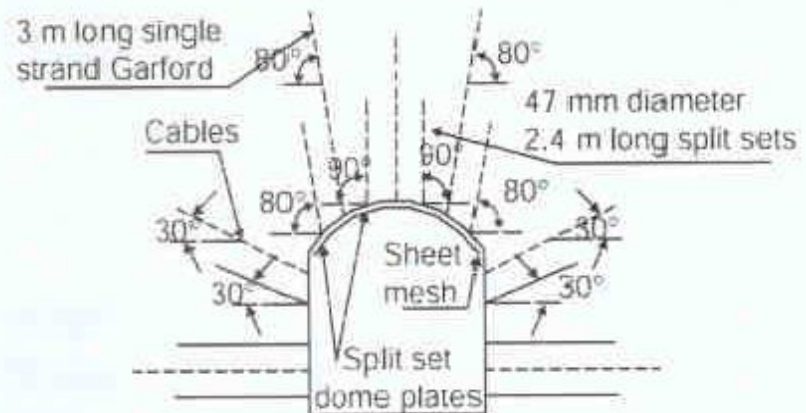
(a) Short term access
(good ground conditions)



(c) Permanent long term access
(average ground conditions)



(b) Permanent long term access
(good ground conditions)



(d) Permanent long term access
(poor ground conditions)

Concrete Support I

- *Concrete is widely used as mine support because of following advantages.
- *High compressive strength.
- *Easy to erect & manufacture.
- *Fire resistance.
- *Smooth finish surface.
- *Suitability under advance mining condition, including presence of abnormal make of water.
- *Problem with concrete are listed.
- *Low tensile strength.
- *Failure with out warning.

Concrete Support II

- *Requirement of curing time for its setting.
- *Concrete support is widely applied in following areas.
- *Shaft lining.
- *Road way lining.
- *Concrete arches.
- *As shot crete or guniting.

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Monolithic Concrete I

- *Monolithic concrete is mass concrete.
- *It is not a concrete block.
- *Monolithic concrete is used to cast a 40 to 60 cm wall around an opening, roof & side.
- *For casting this concrete, shuttering / forms are built & then concrete mixture at ratio of 1:2:4 to 1:1:2 (cement : sand : coarse aggregate etc), depending upon strength required.
- *Concrete structure is allowed to cure for 2 to 4 weeks.
- *Monolithic concrete can further strengthen by incorporating steel bar, rod, straps, angles etc.

Monolithic Concrete II

*These concrete is known as (RCC), reinforced concrete.

*Monolithic concrete is used in shaft lining; filling goes between steel arches & ground (sides & back of mine road ways).

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Shaft Lining Calculation I

Protodjaknov formula

Thickness of lining in shaft

$$t = [(Pr)/\{(c/F)-P\}] + \{150/(c/F)\}$$

t= thickness of lining in cm

P=side pressure on lining in kg/cm sq

r= radius of shaft in cm

c= strength of concrete in kg/cm sq

F= factor of safety, usually 2

Shaft Lining Calculation II

Thickness of lining in shaft

$$t = \{0.007\sqrt{(2rH)}\} + 14$$

$$P = (DH)/(m-1)$$

t= thickness of lining in cm

r= radius of shaft in cm

H= depth of shaft in cm

D= rock density in kg/cm sq

m= poisson ratio.

P=side pressure on lining in kg/cm sq

horizontal stress to shaft lining

Shaft Lining Calculation III

Heber formula

Thickness of shaft lining

$t = [\sqrt{\{(c/F)/(c/F) - 2P\}} - 1]$ up to 400 m depth

$t = [\sqrt{\{(c/F)/(c/F) - \sqrt{3}P\}} - 1]$ more than 400 m depth

t = thickness of lining in cm

c = strength of concrete in kg/cm sq

F = factor of safety, usually 2

P = side pressure on lining in kg/cm sq

Shaft Lining Calculation IV

* t = thickness of lining in cm

* P = side pressure on lining in kg/cm sq

* c = strength of concrete in kg/cm sq

* F = factor of safety, usually 2

* D = rock density in kg/cm sq

* H = depth of shaft in cm

* m = poisson ratio

* r = radius of shaft in cm

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Concrete Block I

- *Arches are built using pre casted concrete blocks.
- *Arches can be made yieldable by inserting wooden piece / block in between concrete block.
- *These wooden block are inserted while carrying out masonry work to build these sets.

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Shotcrete & Guniting I

- *Mortar or concrete conveyed through a hose & pneumatically projected with high velocity on a surface is known as shot creteing.
- *When by this process inly cement mortar is applied & it contains no aggregate, it is called guniting.
- *Guniting has limited application due to its high cost & less effectiveness.
- *There two type of shot crete are available.
- *Dry mixture – mixture of cement & damp sand conveyed through a delivery hose to a nozzle, where remaining water is added.

Shotcrete & Guniting II

- *Dry process equipment has a gun, compressor, hoses, nozzle and water pump (in some design).
- *Wet mix process – all ingredients are mixed before they enter the delivery house.
- *Wet process- there is mixing chamber to which compressed air is fed from mines or through separate supply.
- *These process requires skilled operator.
- *These method used to be applied in a limited space, as such working atmosphere becomes tedious & requires good ventilation & illumination.

Shotcrete & Guniting III

- *Shotcrete is now widely used method for supporting with help of mechanization.
- *Shotcrete is applied by self propelled tier mounted machine.
- *Human exposure to dust has been reduced by providing suitable enclosure in machine.
- *For shotcrete application, surface is cleaned & free from dripping water

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Use of Shotcrete I

- *Shotcrete is applied in shaft, adits, haulage way & service chamber for following purpose.
- *Primary support.
- *Final lining.
- *Protective covering for excavated surface that ore altered, when exposed to air.
- *Protective covering for wooden support, rock bolts etc.
- *As lagging material in place of timber, steel or concrete in between supports.

Use of Shotcrete II

- *Used as rock sealant, this application of shot crete can prevent slaking of shale or other rocks & weathering can be prevented.
- *When an opening is required to be rock bolted, shot crete can be applied before & after rock bolts are installed.
- *When blocky ground tends to produce fragments, it may be applied with wire mesh.
- *Shot crete is usually applied after mesh is bolted to rock surface.
- *Thickness of shot crete varies from 52mm to 104mm (2 inch to 4 inch).

Use of Shotcrete III

- * Shot crete can be reinforced with wire, fiber glass etc.
- * Shot crete is also used to repair stone work, brick work & concrete lining.
- * In making bulk head air tight.
- * Temporary support during shaft sinking.
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Support By Filling I

- *Back fill is next to natural pillar as support.
- *Back fill has 100% ability to support superincumbent load with out yielding.
- *By back filling , worked out area is packed with fill.
- *Fill material can be waste rock, mill tailings, sand etc.
- *Filling is very reliable method of support.
- *Filling is costly process to be executed.
- *Filling allows almost full recovery of ore.
- *Filling decreases other type of support such as timber etc.

Support By Filling II

- *Filling improves fire safety & ventilation.
- *Degree of packing in filling depends on type of filling method & filling material used.
- *Hydraulic filling is wet filling process.
- *Filling material is mixed with water & then transported to filling area through pipe line.
- *Some time cement is added with filling material in wet filling process.
- *Filling is consoled & offer better support also work as artificial pillar in mine.

Support By Filling I

- * Compressed air transportation of filling material through pipe line.
- * Dry filling material (waste rock, mill tailings) can be transported by truck in underground for filling.
- * Ideal filling material should not cake, ignite or oxidized in its loose state.

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Reference I

*Surface and underground Excavations- Methods, Technique and Equipment. Ratan Raj Tatiya, 2005, page 118 to 199.

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